Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is hindered by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts between subjects and sessions. This study proposes SSM-DAN, a domain-adaptive selective state-space network that couples a multi-scale convolutional front-end with bidirectional selective state-space blocks to capture both local and long-range temporal patterns, and reduces subject shift through hierarchical alignment, including a learned source–target normalization layer and label-free test-time adaptation. The framework was evaluated on three public MI-EEG datasets using leave-one-subject-out validation with chronologically separated training, adaptation, and test sessions. The proposed SSM-DAN achieved mean accuracies of 76.12% on BCI Competition IV-2a, 84.09% on BCI Competition IV-2b, and 84.33% on Zhou2016. Ablation studies, which also covered a 54-subject OpenBMI cohort, showed that the contribution of each adaptation component depends on the dataset and that the Riemannian reference should be re-estimated when the signal energy changes between sessions. These results indicate that combining selective state-space modeling with learned, hierarchical alignment improves cross-subject MI decoding without target labels.

Keywords: motor imagery, electroencephalography, brain–computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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1. Introduction

Motor-imagery brain–computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG) for rehabilitation, communication, and control. The event-related desynchronization and synchronization patterns on which they rely are weak, non-stationary, and differ between subjects and between sessions of the same subject [1]. A decoder trained on existing users therefore often degrades on a new one, while calibrating every new user with labeled trials is time-consuming and tiring [2]. Reducing or removing this labeled calibration is thus a central requirement for practical MI-BCIs.

Early MI decoders followed a classical machine-learning pipeline. Common spatial pattern (CSP) filtering [3] and its filter-bank extension FBCSP [4], which won BCI Competition IV, extract band-power features through hand-designed, subject-specific stages and pass them to a linear classifier. Deep learning later replaced these stages with end-to-end feature learning: ShallowConvNet and DeepConvNet [5] and the compact EEGNet [6] learn spectral and spatial filters directly from raw EEG. Later models extended the temporal context: EEG Conformer adds self-attention to a convolutional front-end [7], ATCNet combines attention with dilated temporal convolutional networks (TCNs) [8], and TCFormer joins multi-scale convolution, a Transformer, and a TCN [9]. Self-attention, however, has quadratic cost in sequence length [10] and needs more data than EEG datasets usually provide. More recently, state-space models (SSMs), from the structured S4 [11] to the selective Mamba [12], have emerged as an alternative that scales linearly with sequence length; in language modeling, Mamba matched or exceeded Transformers of comparable size while providing several times higher generation throughput [12]. MI-Mamba and a spatial–temporal Mamba network have since combined SSMs with convolution for MI decoding [13, 14]. Despite their architectural diversity, these approaches are trained on source data and applied unchanged to new users, so their performance depends on how well the training set represents each new user.

A complementary line of work reduces this dependence on the training population and on per-user labeled calibration. EEG foundation models, pretrained on large and diverse corpora such as the roughly 2,500 hours of EEG used by LaBraM [15, 16], are a promising direction, but they require collecting and harmonizing large heterogeneous datasets and are usually fine-tuned with labeled data of the target task. Transfer across subjects of the

same task is a lighter alternative. Domain generalization learns, from several source subjects, features intended to remain valid for unseen subjects, so that a new user can be served without any calibration [17, 18, 19]; because the target subject is never observed, however, its specific shift cannot be compensated. Domain adaptation removes this limitation by also using unlabeled EEG of the new subject. Such data require no trial labels, which makes domain adaptation a practical route to tailoring the decoder to each user without a labeled calibration session. To align source and target features, the maximum mean discrepancy (MMD), introduced into deep networks in its multi-kernel form by DAN [20], is among the most widely used criteria owing to its non-parametric form and stable optimization, and is also adopted by EEG models such as RMETNet [21]. Adversarial training with a domain discriminator [22, 23] and prototype-based alignment [24] provide complementary objectives. Alignment can also be performed inside the network through its normalization layers: in visual domain adaptation, AutoDIAL [25] introduced domain alignment layers that learn how strongly source and target statistics are mixed during normalization, an idea that motivates the feature-level normalization used in the present work.

Beyond the feature level, subject differences are already present in the raw signals: the covariance between EEG channels, which reflects head anatomy and electrode placement, differs markedly across individuals. This suggests aligning the signals themselves before they enter the network. Riemannian geometry, which treats trial covariances as points on the manifold of symmetric positive-definite matrices, has proved effective for MI decoding [26]. Building on it, Zanini et al. proposed Riemannian alignment (RA), which re-centers each subject's covariances at the identity using that subject's label-free Riemannian mean [27]. The resulting accuracy gain, however, can vary across datasets and tends to be more pronounced when the alignment also increases the separation between classes.

We therefore propose SSM-DAN, a selective state-space network with hierarchical, label-free adaptation for cross-subject MI decoding, designed to model the temporal dynamics of MI-EEG efficiently while adapting to each new user without labeled calibration. The main contributions are:

- a decoder that captures both local spectral–spatial patterns and long-range temporal dependencies of MI-EEG at a cost that grows linearly with trial length;
- a hierarchical adaptation strategy that reduces subject shift at the sig-

nal, feature, and decision levels using only unlabeled EEG of the new user; and

- a chronological leave-one-subject-out evaluation on three public datasets, in which target labels are excluded from training, adaptation, and model selection and all competing methods are re-trained under identical conditions.

2. Method

SSM-DAN has two parts (Fig. 1) and is trained in two stages (Fig. 2). An EEG encoder combines multi-scale convolutions for local spectral–spatial features, bidirectional selective state-space blocks for long-range dynamics, and a temporal convolutional fusion head (Sections 2.1–2.3). A hierarchical adaptation scheme reduces subject shift at three levels: Riemannian whitening of the input, AutoDIAL normalization with adversarial and discrepancy losses on the features, and test-time adaptation of the predictions (Sections 2.4–2.7). The model is trained on labeled source-subject trials and unlabeled target-subject trials (Stage 1), adapted to the unlabeled evaluation trials (Stage 2), and then used for prediction (Stage 3).

2.1. Multi-scale convolutional encoder

Each trial $X \in \mathbb{R}^{C\times T}$ (C electrodes, T samples) passes through three temporal convolutions with kernels of 20, 32, and 64 samples, as in TCFormer [9], followed by depthwise spatial filtering, pooling, and grouped refinement to 16 channels per scale. A residual channel-group attention assigns one sigmoid weight to each scale, and a second pooling yields the local feature sequence $H_c \in \mathbb{R}^{48\times L}$.

2.2. Bidirectional selective state-space encoder

To capture trial-wide dependencies, H_c passes through five selective state-space blocks [11, 12], each driving a diagonal recurrence with its projected input u_t:

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \tag{1}$$

$$z_t = C_t^\top h_t + d \odot u_t, \tag{2}$$

where $A = -\exp(A_{\log})$ is a learned stable decay and Δ_t, B_t, and C_t depend on the input, so each block selects which time steps to retain at linear cost.

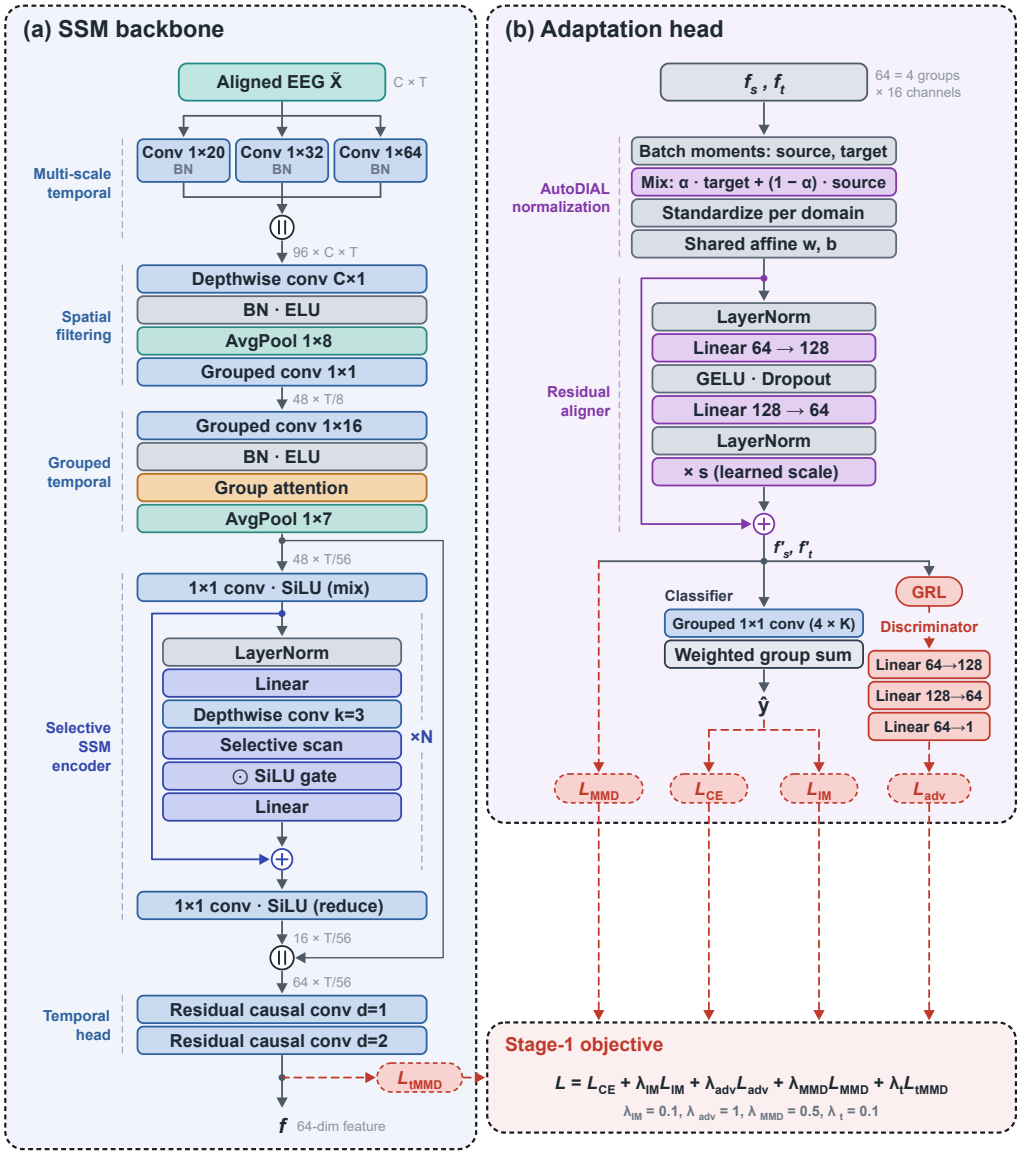


Figure 1: Architecture of SSM-DAN. (a) SSM backbone; tensor sizes are channels × time, $\|$ denotes channel concatenation, $\oplus$ addition, and $\times N$ the $N = 5$ selective SSM blocks with alternating scan direction. (b) Adaptation head: AutoDIAL normalization and a residual aligner feed three parallel branches (MMD, classifier, and GRL with a domain discriminator); their losses and the temporal MMD on the TCN output in (a) form the Stage 1 objective.

Blocks alternate between forward and backward scanning (Fig. 1a), giving bidirectional context without extra blocks. The state dimension is eight, with dropout 0.4 and stochastic depth.

2.3. Local–global fusion and classification

The state-space output is reduced to 16 channels and concatenated with the 48-channel local sequence H_c, so that the head sees both local and global evidence. Two residual grouped temporal convolutional network (TCN) blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The average of its last time step and its temporal mean forms the trial representation $q \in \mathbb{R}^{64}$, made of four 16-channel groups (three convolutional scales and the state-space branch). A norm-constrained grouped classifier produces one set of class scores per group, combined with learned softmax weights.

2.4. Riemannian signal alignment

Because EEG covariance structures are shifted between subjects, Riemannian alignment (RA) whitens each subject's trials by the inverse square root of that subject's mean covariance $\bar{R}$, the affine-invariant Riemannian mean of the OAS-shrunk trial covariances [27, 28]:

$$\widetilde{X} = \bar{R}^{-1/2} X. \tag{3}$$

To avoid leakage, $\bar{R}$ is estimated only from data available before evaluation and then applied unchanged: from the training sessions for the validation sessions of each source subject, and from the unlabeled earlier sessions $\mathcal{T}_a$ for the evaluation sessions $\mathcal{T}_e$ of the target subject.

2.5. Automatic domain alignment layer

Riemannian whitening aligns input statistics but not shifts arising inside the network. Following AutoDIAL [25], the trial representation q is therefore normalized by a layer that learns how strongly target features are standardized with their own statistics: source features use the source moments (μ_s, σ_s^2), and target features use the moments of a mixture of the source and target (μ_t, σ_t^2) distributions,

$$\mu_\alpha = \alpha\mu_t + (1-\alpha)\mu_s, \tag{4}$$

$$\sigma_\alpha^2 = \alpha\sigma_t^2 + (1-\alpha)\sigma_s^2 + \alpha(1-\alpha)(\mu_t - \mu_s)^2, \tag{5}$$

so that

$$\hat{q}^s = \gamma \odot \frac{q^s - \mu_s}{\sqrt{\sigma_s^2 + \epsilon}} + \beta, \qquad \hat{q}^t = \gamma \odot \frac{q^t - \mu_\alpha}{\sqrt{\sigma_\alpha^2 + \epsilon}} + \beta, \tag{6}$$

with shared affine parameters γ and β. The mixing weight $\alpha \in (0, 1)$, initialized to 0.5 and learned separately for each of the four feature groups, interpolates between source statistics ($\alpha \to 0$) and target statistics ($\alpha \to 1$). Moments are computed per mini-batch during training and tracked as running averages (momentum 0.1), which are mixed with the learned α at inference; source validation trials always use source statistics.

2.6. Training objective

A residual $64 \to 128 \to 64$ aligner precedes the classifier and alignment losses. Besides the source cross-entropy $\mathcal{L}_{\text{cls}}$, Stage 1 uses a gradient-reversal domain loss $\mathcal{L}_{\text{adv}}$ [22], MK-MMD losses on trial representations ($\mathcal{L}_{\text{mmd}}$) and TCN sequences ($\mathcal{L}_{\text{tmmd}}$) [20], and the information-maximization loss [29]

$$\mathcal{L}_{\text{IM}} = \frac{1}{N} \sum_{j=1}^{N} H(p_j) - H\left(\frac{1}{N} \sum_{j=1}^{N} p_j \right), \tag{7}$$

where p_j is the predicted class distribution of the j-th of N unlabeled trials and $H(\cdot)$ the entropy. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.1\mathcal{L}_{\text{IM}}^t + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \tag{8}$$

where $\mathcal{L}_{\text{IM}}^t$ is computed on target trials from $\mathcal{T}_a$. The alignment losses see the AutoDIAL output with α fixed, so α is learned only through $\mathcal{L}_{\text{IM}}^t$.

2.7. Test-time adaptation

The checkpoint with the highest source-validation accuracy is adapted to the unlabeled evaluation trials $\mathcal{T}_e$ by information-maximization test-time adaptation (IM-TTA): only the BatchNorm scales and shifts are updated, which limits overfitting to a small target set [30], by minimizing Eq. (7) over $\mathcal{T}_e$ for five Adam passes (learning rate 10^{-4}). The AutoDIAL target moments are then re-estimated in one pass over the same trials, with α and the source moments fixed. The labels of $\mathcal{T}_e$ are used only for scoring. Fig. 2 summarizes both stages.

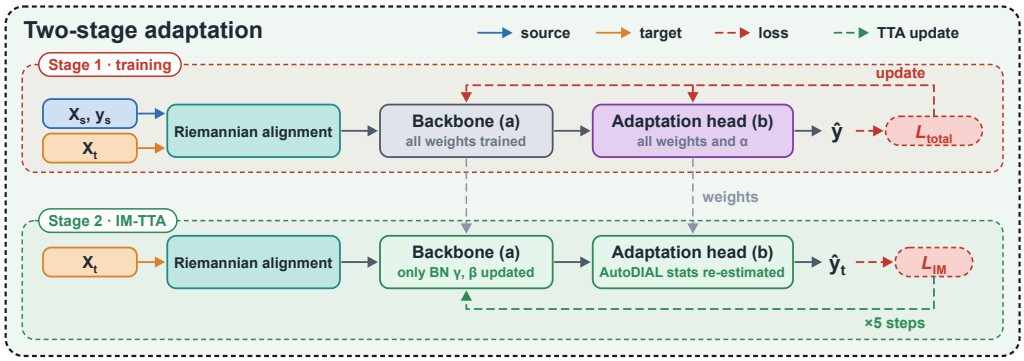


Figure 2: Two-stage adaptation. Stage 1 trains the backbone and the adaptation head (Fig. 1), including α, on labeled source and unlabeled target trials. Stage 2 (IM-TTA) starts from these weights, updates only the backbone BatchNorm parameters γ, β with $\mathcal{L}_{\text{IM}}$ for five steps, and then re-estimates the AutoDIAL target statistics before predicting.

3. Experimental Setup

3.1. Datasets

Three public MI-EEG datasets, accessed through MOABB/Braindecode [31], are used (Table 1).

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Channels	Sessions	Classes
BCI IV-2a	9	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	3	5	left hand, right hand
Zhou2016	4	14	3	left hand, right hand, feet

3.1.1. BCI Competition IV-2a

Nine participants were recorded in two sessions on different days, each with 288 trials of left-hand, right-hand, both-feet, and tongue imagery, from 22 EEG electrodes at 250 Hz (0.5–100 Hz band-pass, 50-Hz notch; the three EOG channels are not used) [32].

3.1.2. BCI Competition IV-2b

Nine right-handed participants, drawn from a study of ten (24.7 ± 3.3 years) [33], performed left- versus right-hand imagery recorded from three

bipolar derivations around C3, Cz, and C4 at 250 Hz (the three EOG channels are not used). Sessions 1–3 form the calibration portion and sessions 4–5 the evaluation portion [32]; this chronological split is kept, and the dataset tests whether the method remains useful with only three electrodes.

3.1.3. Zhou2016

Four healthy participants (22–28 years) were each recorded in three sessions separated by days to months [34]. Each session has two runs of 75 trials (25 per class: left hand, right hand, feet) from 14 electrodes at 250 Hz; each trial has a 5-s imagery cue followed by a 4-s rest.

3.2. Chronological LOSO protocol with test-time adaptation

Each leave-one-subject-out (LOSO) fold holds out one target subject and uses the others as labeled sources, with the session split in Table 2. Earlier target sessions are used without labels in Stage 1; the evaluation sessions are excluded from training and model selection, and their unlabeled EEG is used only by IM-TTA before scoring.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session labels are used only for scoring.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Stage 1 UDA (unlabeled)	IM-TTA + evaluation (labels withheld)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	sessions 1–2	session 3

This gives nine folds for BCI IV-2a and IV-2b and four for Zhou2016, with identical hyperparameters in every fold; the checkpoint with the highest source-validation accuracy is used. For Zhou2016, an all-session target score is reported as an auxiliary diagnostic.

3.3. Preprocessing and augmentation

EEG channels are resampled to 250 Hz and converted to microvolts. BCI IV-2a uses the full MOABB event window and BCI IV-2b drops its last 0.5 s,

both unfiltered; Zhou2016 is band-pass filtered at 8–30 Hz and cropped to 0.5–5.0 s after cue onset.

Z-score statistics are fitted on the source-training partition. For BCI IV-2a and IV-2b, labeled source mini-batches are doubled with the segmentation and reconstruction (S&R) augmentation [2] used by TCFormer [9], which recombines eight-segment pieces of same-class trials; Zhou2016 uses no augmentation.

3.4. Implementation details

All experiments run in Kaggle notebooks on one NVIDIA Tesla T4 GPU (16 GB). BCI IV-2a and IV-2b use a batch size of 48, at most 125 epochs, and Adam (learning rate 9×10^{-4}, $\beta_1 = 0.5$, weight decay 10^{-3}); Zhou2016 uses a batch size of 32, 100 epochs, a learning rate of 10^{-3}, and weight decay 10^{-4}. A three-epoch linear warm-up is followed by cosine decay. The seed (0), data split, and initialization are shared by all methods and variants, so each comparison differs only in the evaluated component.

3.5. Comparison methods

Baselines are grouped by target-data access: source-only EEG Conformer [7], ATCNet [8], TCFormer [9], EEGEncoder [35], and SATrans-Net [36]; UDA-based RMETNet [21]; and test-time methods T-TIME [37] and BFT [38]. Each is compared with the SSM-DAN variant having the same target-data access. All methods are re-trained from released implementations with identical LOSO folds, chronological splits, preprocessing, seed, and source-only checkpoint selection; evaluation labels never enter training or model selection. Model-specific optimization follows the original recommendations. Published results under different protocols are reported separately in Section 4.5. For numerical stability, RMETNet covariances on Zhou2016 receive a ridge of 10^{-3} times their mean eigenvalue.

3.6. Evaluation measures and reporting

Accuracy and Cohen's κ are reported as mean $\pm$ standard deviation across target subjects. Information transfer rate (ITR) [39] expresses accuracy P over N classes as

$$\mathrm{ITR} = \frac{60}{T}\left[\log_2 N + P\log_2 P + (1-P)\log_2\frac{1-P}{N-1}\right] \ \text{bits/min}, \qquad (9)$$

where T is the decoded window (4 s for BCI IV-2a/2b; 4.5 s for Zhou2016); values at or below chance are set to zero before averaging. Per-class recall and macro-F1 assess class balance. Paired accuracy differences use exact two-sided Wilcoxon tests, with Holm correction within each target-access group (Section 3.5); $p_{\text{Holm}} < 0.05$ denotes significance. Efficiency is reported by parameter count, Stage 1 training time, and batch-one CPU latency.

4. Results

Tables 3–5 report subject-wise chronological LOSO results. Standard deviations describe variation across target subjects, not repeated seeds.

4.1. BCI Competition IV-2a

Table 3 reports the four-class results for the nine BCI IV-2a targets.

Table 3: Per-subject LOSO accuracy (%), with mean Cohen's κ and mean information transfer rate (ITR, Eq. (9)), on BCI IV-2a (four classes). p: two-sided exact Wilcoxon signed-rank test against the SSM-DAN model of the same category (Section 3.5); p_{Holm}: p adjusted by the Holm–Bonferroni procedure within each category (Section 3.6). –: reference model.

Method	Target subject accuracy (%)									Mean ± SD			p	p_{Holm}
	S1	S2	S3	S4	S5	S6	S7	S8	S9	Acc. (%)	κ	ITR (bits/min)		
Source-only training														
Conformer [7]	58.68	43.06	78.82	43.06	44.44	41.67	72.92	65.28	49.31	55.25 ± 13.39	0.403 ± 0.179	5.29 ± 4.58	0.0039	0.0312
ATCNet [8]	69.44	40.97	76.74	50.35	43.40	42.01	61.81	70.49	57.99	57.02 ± 12.74	0.427 ± 0.170	5.73 ± 4.19	0.0039	0.0312
TCFormer [9]	65.28	49.65	79.86	54.86	61.81	38.89	71.88	66.67	58.33	60.80 ± 11.49	0.477 ± 0.153	6.80 ± 4.02	0.0039	0.0312
EEGEncoder [35]	64.24	49.31	78.82	46.18	54.86	40.97	74.31	65.97	47.92	58.06 ± 12.55	0.441 ± 0.167	6.04 ± 4.43	0.0039	0.0312
SATrans-Net [36]	68.06	47.92	73.96	53.47	61.11	45.14	73.61	71.53	63.19	62.00 ± 10.34	0.493 ± 0.138	7.09 ± 3.62	0.0039	0.0312
SSM-DAN (source-only)	68.06	51.04	86.46	61.11	64.24	39.58	72.92	73.61	62.15	64.35 ± 12.77	0.525 ± 0.170	8.22 ± 4.97	0.0039	0.0312
Unsupervised domain adaptation														
RMETNet [21]	67.01	39.24	77.78	55.90	49.65	52.78	63.54	66.67	68.40	60.11 ± 11.05	0.468 ± 0.147	6.52 ± 3.73	0.0039	0.0039
SSM-DAN (Stage 1 UDA)	79.17	47.57	92.71	73.61	57.64	59.72	82.29	82.64	73.96	72.15 ± 13.58	0.629 ± 0.181	11.60 ± 6.31	0.0039	0.0312
Test-time adaptation														
T-TIME [37]	67.01	53.47	78.47	64.58	68.75	49.65	79.51	69.44	65.28	66.24 ± 9.34	0.550 ± 0.125	8.56 ± 3.73	0.0742	0.0742
BFT [38]	64.24	44.44	76.74	65.62	61.81	51.39	73.61	70.49	59.38	63.08 ± 9.77	0.508 ± 0.130	7.42 ± 3.51	0.0039	0.0078
SSM-DAN (IM-TTA)	77.43	47.92	88.19	66.32	64.24	61.81	79.86	82.99	73.61	71.37 ± 11.78	0.618 ± 0.157	10.98 ± 5.27	0.0039	0.0312
SSM-DAN	**79.51**	**52.43**	**93.40**	**73.96**	**69.44**	**65.97**	**88.89**	**86.11**	**75.35**	**76.12 ± 11.96**	**0.682 ± 0.159**	**13.30 ± 6.15**	–	–

SSM-DAN obtained the highest accuracy and κ, and significantly outperformed every source-only baseline and its own source-only variant on all nine targets ($p_{\text{Holm}} = 0.0312$). IM-TTA improved every target over SSM-DAN (Stage 1 UDA). In the UDA group, SSM-DAN (Stage 1 UDA) significantly exceeded RMETNet ($60.11 \pm 11.05\%$) on all nine targets. In the test-time adaptation group, SSM-DAN (IM-TTA) significantly exceeded BFT on all nine targets and exceeded T-TIME on seven, although the latter difference was not significant.

4.2. BCI Competition IV-2b

Table 4 reports the two-class results for the nine BCI IV-2b targets, which provide only three bipolar EEG derivations.

Table 4: Per-subject LOSO accuracy (%), with mean Cohen's κ and mean information transfer rate (ITR, Eq. (9)), on BCI IV-2b (two classes). p: two-sided exact Wilcoxon signed-rank test against the SSM-DAN model of the same category (Section 3.5); p_{Holm}: p adjusted by the Holm–Bonferroni procedure within each category (Section 3.6). –: reference model.

Method	Target subject accuracy (%)									Mean ± SD			p	p_{Holm}
	S1	S2	S3	S4	S5	S6	S7	S8	S9	Acc. (%)	κ	ITR (bits/min)		
Source-only training														
Conformer [7]	75.31	62.86	55.62	92.50	94.69	74.06	82.81	82.50	81.88	78.03 ± 12.02	0.561 ± 0.240	4.55 ± 3.51	0.0156	0.1094
ATCNet [8]	76.25	65.71	57.19	90.00	96.88	77.50	84.06	84.69	82.19	79.38 ± 11.40	0.588 ± 0.228	4.89 ± 3.58	0.0078	0.0625
TCFormer [9]	78.44	69.64	64.38	90.62	95.00	75.94	86.87	85.31	84.38	81.18 ± 9.37	0.624 ± 0.187	5.17 ± 3.16	0.0820	0.3281
EEGEncoder [35]	73.44	65.71	59.06	92.50	97.19	85.00	85.94	86.56	83.75	81.02 ± 11.76	0.620 ± 0.235	5.48 ± 3.81	0.0547	0.2734
SATrans-Net [36]	77.50	69.29	69.69	90.62	97.50	79.69	87.50	84.38	79.69	81.76 ± 8.78	0.635 ± 0.176	5.36 ± 3.45	0.2109	0.6094
SSM-DAN (source-only)	77.81	68.57	63.13	89.38	96.56	73.44	84.69	86.25	85.00	80.54 ± 10.05	0.611 ± 0.201	5.07 ± 3.44	0.0156	0.1094
Unsupervised domain adaptation														
RMETNet [21]	73.12	57.86	55.00	91.56	94.06	82.81	85.62	88.75	83.13	79.10 ± 13.39	0.582 ± 0.268	5.04 ± 3.54	0.0117	0.0117
SSM-DAN (Stage 1 UDA)	78.44	70.71	63.44	90.31	97.81	83.75	89.69	95.63	85.94	83.97 ± 10.71	0.679 ± 0.214	**6.42 ± 3.98**	1.0000	1.0000
Test-time adaptation														
T-TIME [37]	59.69	65.36	85.31	93.75	97.19	85.00	74.69	91.25	59.06	79.03 ± 13.94	0.581 ± 0.279	5.24 ± 4.39	0.3125	0.3125
BFT [38]	75.00	65.00	59.69	93.44	94.69	79.06	88.12	94.37	81.25	81.18 ± 12.12	0.624 ± 0.242	5.60 ± 3.96	0.0078	0.0156
SSM-DAN (IM-TTA)	77.81	68.57	66.25	93.75	96.25	81.88	89.69	94.69	81.25	83.35 ± 10.50	0.667 ± 0.210	6.15 ± 3.92	0.2031	0.6094
SSM-DAN	77.81	69.29	65.94	90.00	97.19	84.69	91.56	95.31	85.00	**84.09 ± 10.42**	**0.682 ± 0.208**	**6.42 ± 3.90**	–	–

SSM-DAN again obtained the highest mean accuracy. It exceeded its source-only variant on all seven targets with non-tied accuracies, ATCNet on all eight, and EEG Conformer on eight of nine ($p = 0.0078$–0.0156), but these differences did not remain significant after Holm correction ($p_{\text{Holm}} = 0.0625$–0.1094); its advantages over TCFormer, EEGEncoder, SATrans-Net, and SSM-DAN (Stage 1 UDA) were small and not significant, with IM-TTA helping on only three targets. Within the matched groups, SSM-DAN (Stage 1 UDA) significantly exceeded RMETNet and SSM-DAN (IM-TTA) significantly exceeded BFT; SSM-DAN (IM-TTA) also exceeded T-TIME ($79.03 \pm 13.94\%$) on average, but not significantly, as T-TIME varied far more across targets.

4.3. Zhou2016

Table 5 reports the three-class results on target session 3 of Zhou2016. With four folds, the smallest attainable two-sided Wilcoxon p-value is 0.125, so no difference can be significant.

Table 5: Per-subject LOSO accuracy (%), with mean Cohen's κ and mean information transfer rate (ITR, Eq. (9)), on session 3 of Zhou2016 (three classes). Bold: best per column. p: two-sided exact Wilcoxon signed-rank test against the SSM-DAN model of the same category (Section 3.5); p_{Holm}: p adjusted by the Holm–Bonferroni procedure within each category (Section 3.6). –: reference model.

Method	Target subject accuracy (%)				Mean ± SD			p	p_{Holm}
	S1	S2	S3	S4	Acc. (%)	κ	ITR (bits/min)		
Source-only training									
Conformer [7]	76.00	79.33	70.67	82.67	77.17 ± 4.43	0.657 ± 0.066	7.86 ± 1.86	0.1250	1.0000
ATCNet [8]	72.00	82.67	76.67	74.67	76.50 ± 3.93	0.648 ± 0.059	7.60 ± 1.70	0.1250	1.0000
TCFormer [9]	68.00	78.67	66.67	80.67	73.50 ± 6.22	0.603 ± 0.093	6.67 ± 2.39	0.1250	1.0000
EEGEncoder [35]	62.00	84.00	74.67	76.67	74.33 ± 7.92	0.615 ± 0.119	7.07 ± 2.98	0.2500	1.0000
SATrans-Net [36]	81.33	80.00	67.33	70.00	74.67 ± 6.09	0.620 ± 0.091	7.06 ± 2.40	0.1250	1.0000
SSM-DAN (source-only)	72.00	78.67	71.33	78.67	75.17 ± 3.51	0.627 ± 0.053	7.10 ± 1.41	0.1250	1.0000
Unsupervised domain adaptation									
RMETNet [21]	70.00	**89.33**	80.67	84.67	81.17 ± 7.14	0.718 ± 0.107	9.63 ± 3.29	0.8750	1.0000
SSM-DAN (Stage 1 UDA)	79.33	84.67	**87.33**	78.00	82.33 ± 3.82	0.735 ± 0.057	9.90 ± 1.92	0.6250	1.0000
Test-time adaptation									
T-TIME [37]	**88.67**	84.00	81.33	**86.00**	$\mathbf{85.00 \pm 2.69}$	$\mathbf{0.775 \pm 0.040}$	$\mathbf{11.06 \pm 1.46}$	0.5000	0.8750
BFT [38]	84.00	86.67	76.00	84.67	82.83 ± 4.07	0.743 ± 0.061	10.13 ± 1.94	0.8750	1.0000
SSM-DAN (IM-TTA)	79.33	84.00	83.33	83.33	82.50 ± 1.85	0.738 ± 0.028	9.90 ± 0.89	0.2500	1.0000
SSM-DAN	84.00	83.33	84.67	85.33	84.33 ± 0.75	0.765 ± 0.011	10.70 ± 0.39	–	–

SSM-DAN had the most consistent performance across targets and exceeded every source-only baseline and its source-only variant on all four targets, except EEGEncoder, which it exceeded on three. In the UDA group, RMETNet was slightly below its reference SSM-DAN (Stage 1 UDA) on average and far more variable across targets, although it obtained the highest accuracy on target 2. In the test-time adaptation group, T-TIME obtained the highest mean accuracy on this dataset, above both its reference SSM-DAN (IM-TTA) and the complete SSM-DAN, while BFT was on par with the reference. SSM-DAN also remained above TCFormer in the auxiliary score pooled over all three target sessions.

4.4. Information transfer rate and class-wise performance

In practical terms, the accuracy gains of SSM-DAN translate into faster communication. On BCI IV-2a, adaptation raised its ITR by about 60% over the source-only variant (13.3 versus 8.2 bits/min; Table 3), so a new user could issue markedly more commands per minute without any labeled calibration. On BCI IV-2b and Zhou2016 the adaptive methods operated in a similar range, mirroring the accuracy results.

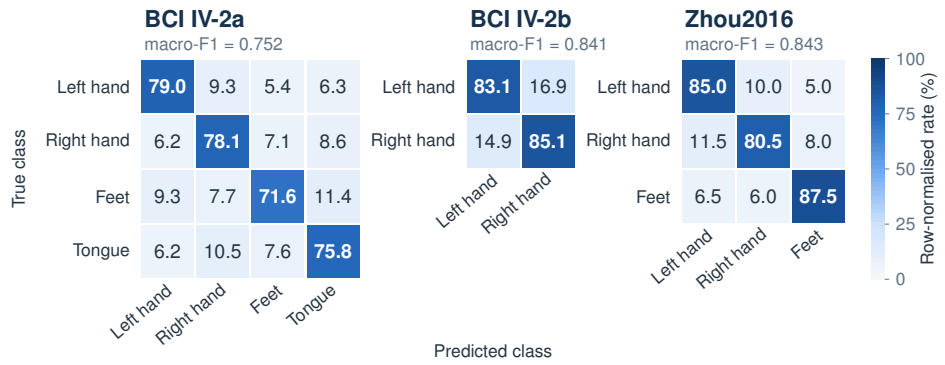


Figure 3: Mean confusion matrices of SSM-DAN on the evaluation sessions of BCI IV-2a (left), BCI IV-2b (middle), and Zhou2016 (right). Each subject's matrix is normalized by the number of trials of each true class and then averaged over target subjects, so the diagonal gives the mean recall of each class; the macro-averaged F1 score is given under each dataset name.

Fig. 3 shows that these gains are spread across classes rather than obtained by favoring an easy one: macro-F1 tracks accuracy on every dataset and no class is systematically neglected. This matters for test-time adaptation, whose entropy objective could otherwise collapse predictions onto a few classes. The remaining errors are also physiologically plausible: on BCI IV-2a, feet imagery was most often confused with tongue imagery, two tasks represented near the midline of the sensorimotor cortex, whereas the lateralized hand classes were recognized best.

4.5. Comparison with published results

Table 6 contrasts the cross-subject accuracies reported in the original articles with the re-runs of this study. The published values are not directly comparable, because the protocols differ in four respects. (i) *Task*: the T-TIME and BFT articles report two-class left- versus right-hand MI, whereas BCI IV-2a and Zhou2016 are evaluated here with four and three classes. (ii) *Session shift and target data*: several protocols train and test within the same recording session, and the EEGEncoder article reports no cross-subject result at all, its multi-subject figure being obtained with the test subjects' own first sessions in the training set; here every target is unseen during training and scored on a later session recorded on a different day. (iii) *Model selection*: the released scripts of SATrans-Net and RMETNet report the highest, or the mean of the five highest, per-epoch accuracies on the held-out subject, i.e.,

the evaluation labels select the epoch; here the checkpoint is chosen on source data only. (iv) *Transductive access*: RMETNet uses the held-out evaluation trials themselves as the target domain of its MMD loss and Riemannian alignment, whereas here domain adaptation sees only the earlier, unlabeled target sessions. Each of (ii)–(iv) makes a protocol easier or its estimate more optimistic, which largely explains why most re-runs are lower on BCI IV-2a. Where the split matches ours, the gaps are small: the TCFormer benchmark [9] uses the same session split and differs only in averaging the last-epoch model over five seeds and in model-specific preprocessing, and the re-runs of TCFormer and ATCNet lie within three points of it on both BCI datasets, while EEG Conformer is higher here. On BCI IV-2b, where the shift between calibration and evaluation sessions is smaller, the re-runs reach or exceed several published values (e.g., SATrans-Net, 81.76 vs. 79.64%).

Table 6: Published cross-subject accuracy (%) of the baselines versus their re-runs under the protocol of this study (Table 2). –: not reported; settings (a)–(e) are given below the table.

		Published			This study (chronological LOSO)		
Method	Setting	BCI IV-2a	BCI IV-2b	Zhou2016	BCI IV-2a	BCI IV-2b	Zhou2016
EEG Conformer [7]	(a) [9]	45.59	75.25	–	55.25	78.03	77.17
ATCNet [8]	(a) [9]	59.66	80.94	–	57.02	79.38	76.50
TCFormer [9]	(a)	63.00	81.34	–	60.80	81.18	73.50
EEGEncoder [35]	(b)	–	–	–	58.06	81.02	74.33
SATrans-Net [36]	(c)	68.13	79.64	–	62.00	81.76	74.67
RMETNet [21]	(d)	71.39	80.93	–	60.11	79.10	81.17
T-TIME [37]	(e)	79.30†	–	–	66.24	79.03	85.00
BFT [38]	(e)	77.47†	–	84.38†	63.08	81.18	82.83

(a) Same session split as here, last-epoch model averaged over five seeds, model-specific preprocessing. (b) No cross-subject result; the 74.48% on BCI IV-2a trains on session 1 of all subjects, including the test subject. (c) LOSO without a reported session split, best per-epoch test accuracy. (d) One session per subject, held-out subject used as MMD/RA target, mean of the five best epochs. (e) Online test-time adaptation on session 1 only. †Two-class (left- vs. right-hand) task.

4.6. Ablation study

Table 7 follows the primary chronological protocol. The variant without domain adaptation removes every Stage 1 domain-adaptation component (AutoDIAL, target information maximization, and the adversarial and both MMD losses) while retaining Riemannian alignment and IM-TTA; all variants otherwise use the same folds and checkpoint rule.

OpenBMI (Lee2019_MI) [40] is added as a larger cohort for the ablation. It contains 54 subjects recorded with 62 EEG channels in two sessions on different days, each with one offline run of 100 left- versus right-hand MI trials. Signals are band-pass filtered at 4–38 Hz, re-referenced to the common average, downsampled to 100 Hz, and cut to the 0–4-s cue interval; S&R augmentation is not used. The two sessions allow the chronological protocol of the other datasets: in each of the 54 LOSO folds, the 53 source subjects provide labeled session-1 trials for training and session-2 trials for checkpoint selection, while the target's unlabeled session 1 is used in Stage 1 and its session 2 for IM-TTA and evaluation. RA is fitted on session 1 of each subject, and all variants use the BCI IV-2a optimization settings. Because every evaluation session has 100 class-balanced trials, $\kappa = 2P-1$ for accuracy P. With 54 paired targets, the OpenBMI Wilcoxon tests use the normal approximation instead of exact enumeration, with Holm correction over the four ablation comparisons.

Table 7: Ablation results: mean ± standard deviation of accuracy (%) and Cohen's κ across target subjects. The best variant of each dataset is in bold. The last column replaces the fixed RA reference of SSM-DAN by a session-specific one (Section 4.7); –: not evaluated. [a]53 and [b]47 of the 54 OpenBMI targets.

Dataset	Metric	SSM-DAN	Source-only	Without Riemannian alignment	Without domain adaptation	Without test-time adaptation	Session-specific RA reference
BCI IV-2a	Acc.	76.12 ± 11.96	64.35 ± 12.77	70.29 ± 11.60	71.37 ± 11.78	72.15 ± 13.58	$\mathbf{77.74 \pm 11.72}$
	κ	0.682 ± 0.159	0.525 ± 0.170	0.604 ± 0.155	0.618 ± 0.157	0.629 ± 0.181	$\mathbf{0.703 \pm 0.156}$
BCI IV-2b	Acc.	84.09 ± 10.42	80.54 ± 10.05	83.58 ± 10.46	83.35 ± 10.50	83.97 ± 10.71	$\mathbf{84.10 \pm 10.23}$
	κ	$\mathbf{0.682 \pm 0.208}$	0.611 ± 0.201	0.672 ± 0.209	0.667 ± 0.210	0.679 ± 0.214	$\mathbf{0.682 \pm 0.205}$
Zhou2016	Acc.	$\mathbf{84.33 \pm 0.75}$	75.17 ± 3.51	84.00 ± 3.16	82.50 ± 1.85	82.33 ± 3.82	–
	κ	$\mathbf{0.765 \pm 0.011}$	0.627 ± 0.053	0.760 ± 0.047	0.738 ± 0.028	0.735 ± 0.057	–
OpenBMI	Acc.	80.61 ± 10.26	79.92 ± 11.04^{a}	83.50 ± 9.43	79.85 ± 10.02	76.65 ± 12.19	$\mathbf{84.21 \pm 9.70}^{b}$
	κ	0.612 ± 0.205	0.598 ± 0.221^{a}	0.670 ± 0.189	0.597 ± 0.200	0.533 ± 0.244	$\mathbf{0.684 \pm 0.194}^{b}$

On BCI IV-2a, removing Riemannian alignment, Stage 1 domain adaptation, or test-time adaptation each significantly reduced accuracy. On BCI IV-2b and Zhou2016, the complete model remained the best variant, but its margins were small and not significant. The contribution of each component is therefore dataset dependent. On OpenBMI, IM-TTA increased accuracy by 3.96 points (higher for 38 of 54 targets, $p_{\text{Holm}} < 0.001$), whereas the gains over the variant without domain adaptation (0.76 points) and over the

source-only model (0.51 points) were not significant ($p_{\text{Holm}} = 0.68$ and 0.72). Removing RA, however, increased accuracy by 2.89 points (higher for 33 of 54 targets, $p_{\text{Holm}} = 0.002$): with a reference fixed on session 1, RA reduced accuracy on this dataset. Section 4.7 examines this effect.

4.7. Session-specific RA reference

The fixed reference assumes that a subject's mean covariance is stable across sessions. On OpenBMI this assumption fails: after session-1 RA, the median trial energy of session 2 was 1.55 times that of session 1 across subjects, and more than twice for 20 of the 54 subjects, compared with 1.08–1.39 on BCI IV-2a. SSM-DAN was therefore re-run with a session-specific reference, in which every session of every subject, including the target evaluation session, is whitened by the Riemannian mean of its own unlabeled trials. The evaluation-session reference uses the same unlabeled evaluation trials that IM-TTA already accesses, so the target-data access is unchanged; like IM-TTA, it needs these trials as a batch. All other settings are those of the main experiments; the results are in the last column of Table 7.

On OpenBMI, for the 47 targets completed with both references, the session-specific reference added 3.47 points with IM-TTA (84.21% vs. 80.74%; higher for 38 of 47 targets, $p < 0.001$) and 7.62 points without it (84.00% vs. 76.38%; 40 of 47, $p < 0.001$), while the gain of IM-TTA fell from 4.36 to 0.21 points ($p = 0.32$); most of the IM-TTA gain under the fixed reference therefore compensated the energy shift between sessions. With the session-specific reference, SSM-DAN also exceeded the variant without RA on the same targets (84.21% vs. 83.38%), although not significantly ($p = 0.35$). On BCI IV-2a, accuracy rose by 1.62 points with IM-TTA (8 of 9 targets, exact $p = 0.023$), and IM-TTA still added 2.89 points (74.85% to 77.74%; 8 of 9, $p = 0.008$). On BCI IV-2b, the two references gave the same accuracy (84.10% vs. 84.09%; higher for 3 targets, lower for 5, $p = 0.97$). Because this variant was examined after the main experiments, these tests are not Holm-corrected and serve as supporting evidence; all other results use the fixed reference.

4.8. Computational efficiency

Table 8 reports trainable parameters, Stage 1 training time, and CPU response time. The source-only variant is smaller because it omits the AutoDIAL layer, aligner, and domain discriminator; RA and IM-TTA do not change parameter count.

Stage 1 adaptation accounts for most of the extra training time of SSM-DAN over its source-only variant (nearly doubling it on Zhou2016), whereas IM-TTA adds little. SSM-DAN has a parameter count comparable to ATC-Net and TCFormer and far below EEG Conformer, while T-TIME and BFT are the smallest and fastest methods.

Table 8: Computational cost: trainable parameters, mean Stage 1 training time per fold, and single-trial CPU response time. SSM-DAN variants as in Section 3.5.

	BCI IV-2a			BCI IV-2b			Zhou2016		
Method	Params (K)	Train (min)	CPU (ms)	Params (K)	Train (min)	CPU (ms)	Params (K)	Train (min)	CPU (ms)
Source-only training									
Conformer	789.6	11.14	11.45	759.1	4.93	9.34	858.7	1.53	10.34
ATCNet	113.7	48.58	13.33	112.8	60.19	11.35	113.3	8.09	11.91
TCFormer	127.2	38.37	14.96	123.4	31.04	11.17	125.6	5.89	15.97
EEGEncoder	197.2	13.49	26.51	196.2	14.98	26.88	196.8	3.64	25.68
SATrans-Net	155.6	10.19	12.78	153.7	4.99	11.10	155.0	1.32	11.38
SSM-DAN (source-only)	98.4	47.01	25.08	94.6	36.10	13.68	96.8	6.29	13.36
Unsupervised domain adaptation									
RMETNet	193.8	45.88	11.15	159.1	21.96	4.90	306.6	23.74	7.19
SSM-DAN (Stage 1 UDA)	132.0	58.45	18.44	128.2	48.08	13.35	130.4	11.42	14.65
Test-time adaptation									
T-TIME	9.5	5.85	7.71	6.3	3.07	5.08	8.4	1.34	6.72
BFT	40.6	1.47	3.08	39.9	0.96	2.61	51.0	0.36	2.84
SSM-DAN (IM-TTA)	131.9	40.73	22.12	128.1	33.21	14.25	130.3	6.46	18.41
SSM-DAN (w/o RA)	132.0	58.41	21.63	128.2	47.95	13.33	130.4	11.66	18.88
SSM-DAN	**132.0**	**59.82**	**22.20**	**128.2**	**54.21**	**17.40**	**130.4**	**11.70**	**19.14**

5. Discussion

The benefit of each adaptation component is dataset dependent: it was clear on BCI IV-2a but small and not significant on BCI IV-2b and Zhou2016, where repeated-seed experiments are needed to confirm the differences. The larger OpenBMI cohort confirmed the gain of IM-TTA but showed that an RA reference fixed on an earlier session can reduce accuracy when the signal energy changes between sessions. Re-estimating the reference on each session removed this loss and accounted for most of the IM-TTA gain, which suggests that a large part of the session shift handled by test-time adaptation is a change of second-order signal statistics.

The alternating scans exploit complete offline trials and are non-causal, so online use would require forward-only processing.

Because IM-TTA, and the session-specific RA reference, use unlabeled evaluation-session signals, the method is transductive and must be compared with baselines of identical target access; online use would require running estimates of the reference and of the adapted statistics. Further limitations are the three-channel BCI IV-2b setting, the small Zhou2016 cohort, and incomplete matched baselines.

6. Conclusion

We presented SSM-DAN, a cross-subject MI-EEG framework that combines multi-scale convolution and bidirectional selective state-space modeling with hierarchical UDA and constrained IM-TTA, evaluated under a chronological LOSO protocol that keeps target labels out of training, adaptation, and model selection. It reached mean accuracies of 76.12% on BCI IV-2a, 84.09% on BCI IV-2b, and 84.33% on Zhou2016. On the 54-subject OpenBMI cohort, the ablation showed that the RA reference should be re-estimated on each session when the signal energy drifts between sessions. Matched baselines and repeated-seed experiments are still needed before drawing broader conclusions.

CRediT authorship contribution statement

[Author name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All EEG data analyzed in this study are publicly available. BCI Competition IV datasets 2a and 2b [32], the Zhou2016 dataset [34], and the OpenBMI dataset [40] were accessed through the MOABB library [31]; no new data were collected.

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